

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	138781	46.075111
1	[Git, Change, Create]	44929	14.916370
2	[Git, Change, Delete]	22781	7.563262
3	[Java, ByteCode, Method, Member]	13516	4.487294
4	[Java, ByteCode, Parameter]	13306	4.417575
5	[Git, Commit]	11390	3.781465
6	[File, Git]	8286	2.750941
7	[Java, ByteCode, ParameterizedType, Bound]	7286	2.418943
8	[Java, ByteCode, Bound]	7245	2.405331
9	[Git, Change, Rename]	7018	2.329967
10	[Java, ByteCode, Member, Field]	3605	1.196855
11	[Java, ByteCode, Bound, WildcardType]	2950	0.979396
12	[Java, Value, ByteCode, Annotation]	2936	0.974748
13	[Xml, Element]	2162	0.717781
14	[Java, ByteCode, Constructor, Method, Member]	2109	0.700185
15	[Xml, Text]	1450	0.481398
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.368851
17	[Java, ByteCode, Method, Member, Lambda]	968	0.321375
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	890	0.295479
19	[Type, File, Java, Class, ByteCode]	845	0.280539
20	[Json, Key]	739	0.245347
21	[Value, Json, Scalar]	723	0.240035
22	[Java, Value, ByteCode, Primitive]	673	0.223435
23	[Type, File, Java, ByteCode, JavaType]	649	0.215467
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.191895
25	[Type, File, Java, ByteCode, ExternalType]	412	0.136783
26	[Git, Change, Copy]	298	0.098936
27	[Author, Git, Person]	295	0.097940
28	[Value, Array]	289	0.095948
29	[Committer, Git, Person]	242	0.080344

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

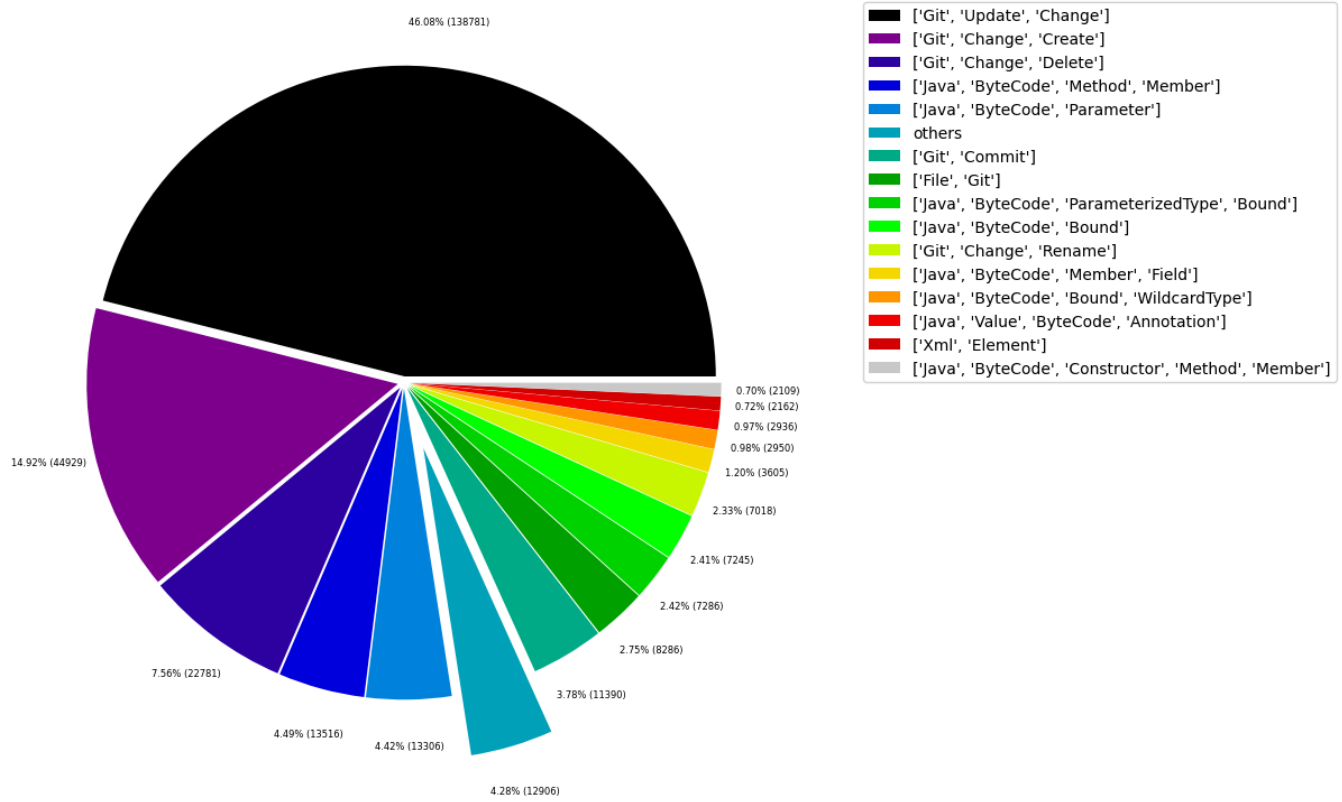


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000664
4	[File]	3	0.000996
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001328
6	[Maven, Exclusion]	5	0.001660
7	[Value, Array, Json]	6	0.001992
8	[File, Maven, Xml, Pom, Document]	9	0.002988
9	[Type, File, Java, ByteCode, Void]	9	0.002988
10	[Java, ManifestSection]	9	0.002988
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002988
12	[File, Java, Manifest]	9	0.002988
13	[File, Java, ServiceLoader]	10	0.003320
14	[File, Java, Properties]	12	0.003984
15	[Maven, ExecutionGoal]	16	0.005312
16	[Maven, PluginExecution]	16	0.005312
17	[Xml, Attribute]	18	0.005976
18	[JQAssistant, Rule, Concept]	19	0.006308
19	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006972
20	[Maven, Plugin]	21	0.006972
21	[Maven, Configuration]	21	0.006972
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007304
23	[Type, File, Java, ByteCode, Enum]	28	0.009296
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009960
25	[Xml, Namespace]	36	0.011952
26	[Type, File, Java, ByteCode, Annotation]	44	0.014608
27	[File, Directory]	44	0.014608
28	[Type, File, Java, Class, ByteCode, Throwable]	55	0.018260
29	[Type, File, Java, ByteCode, Interface, Generi...	85	0.028220

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

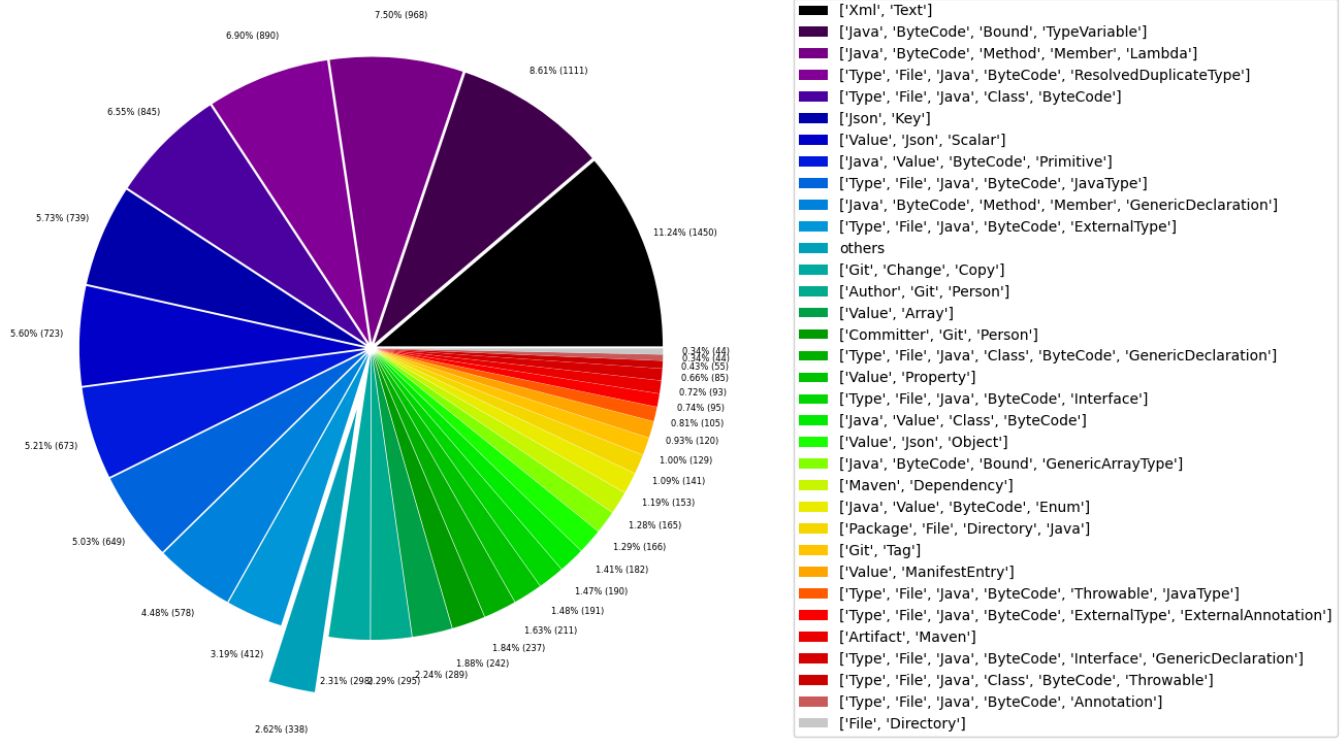


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

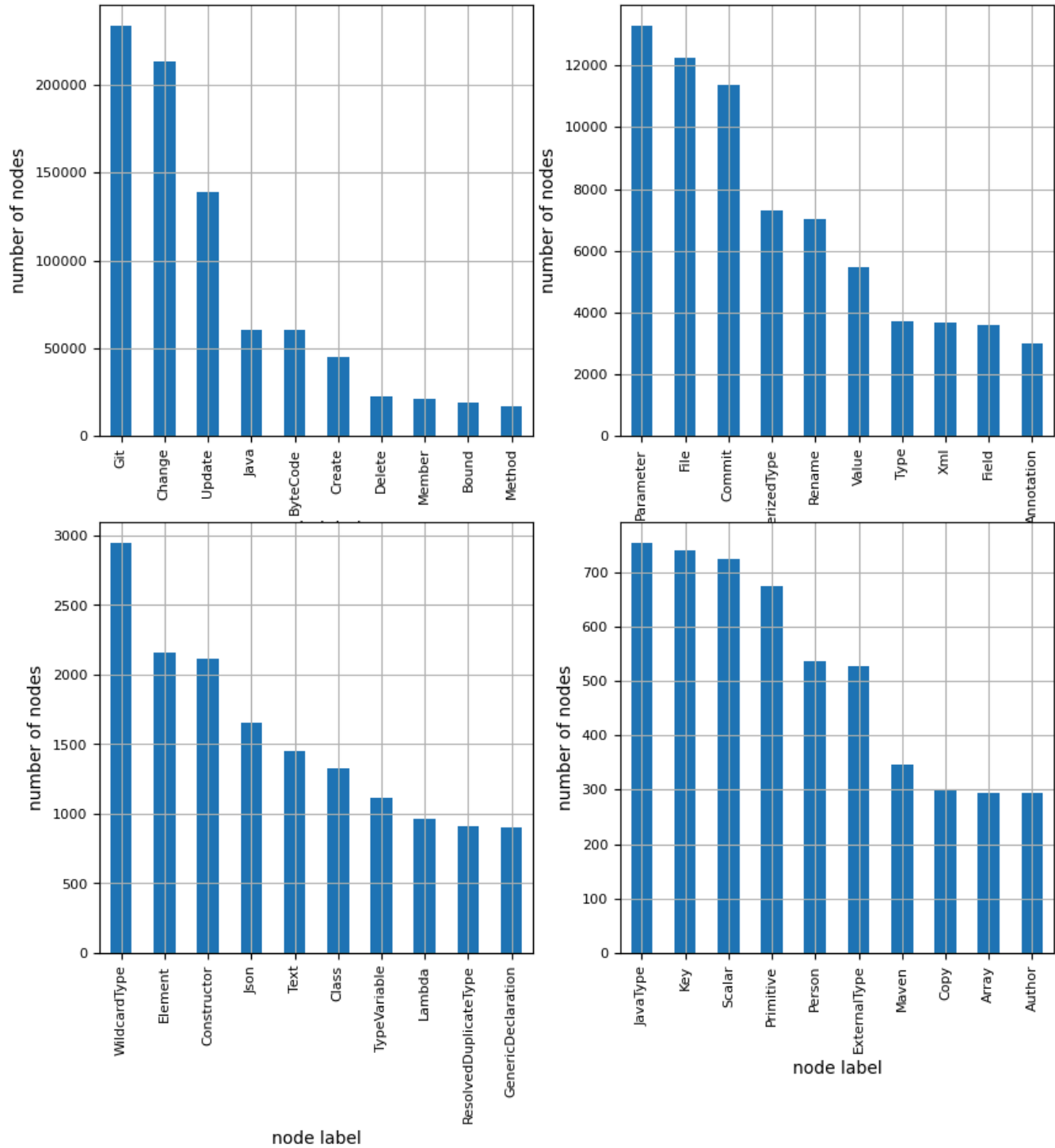
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	234151	77.737827
1	Change	213807	70.983646
2	Update	138781	46.075111
3	Java	60704	20.153649
4	ByteCode	60514	20.090569
5	Create	44929	14.916370
6	Delete	22781	7.563262
7	Member	20780	6.898933
8	Bound	18758	6.227632
9	Method	17175	5.702078
10	Parameter	13306	4.417575
11	File	12244	4.064992
12	Commit	11390	3.781465
13	ParameterizedType	7286	2.418943
14	Rename	7018	2.329967
15	Value	5483	1.820349
16	Type	3718	1.234371
17	Xml	3675	1.220095
18	Field	3605	1.196855
19	Annotation	2980	0.989356
20	WildcardType	2950	0.979396
21	Element	2162	0.717781
22	Constructor	2113	0.701513
23	Json	1652	0.548462
24	Text	1450	0.481398
25	Class	1327	0.440562
26	TypeVariable	1111	0.368851
27	Lambda	968	0.321375
28	ResolvedDuplicateType	912	0.302783
29	GenericDeclaration	904	0.300127
30	JavaType	754	0.250327
31	Key	739	0.245347
32	Scalar	723	0.240035
33	Primitive	673	0.223435
34	Person	537	0.178283
35	ExternalType	528	0.175295
36	Maven	346	0.114872
37	Copy	298	0.098936
38	Array	295	0.097940
39	Author	295	0.097940

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 935709

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213807	22.849732
1	MODIFIES	213807	22.849732
2	UPDATES	138781	14.831641
3	CREATES	52245	5.583467
4	INVOKES	36433	3.893625
5	DELETES	29799	3.184644
6	COMMITTED	22780	2.434518
7	DEPENDS_ON	22504	2.405021
8	OF_TYPE	21945	2.345280
9	DECLARES	21240	2.269936
10	OF_RAW_TYPE	17367	1.856026
11	HAS	14444	1.543642
12	HAS_PARENT	13921	1.487749
13	RETURNS	12858	1.374145
14	HAS_COMMIT	11390	1.217259
15	READS	9382	1.002662
16	HAS_ACTUAL_TYPE_ARGUMENT	8414	0.899211
17	HAS_FILE	8286	0.885532
18	RENAMES	7018	0.750020
19	OF_GENERIC_TYPE	6000	0.641225
20	RESOLVES_TO	4188	0.447575
21	SIMILAR	4078	0.435819
22	WRITES	3948	0.421926
23	CONTAINS	3928	0.419789
24	HAS_NEW_NAME	3662	0.391361
25	RETURNS_GENERIC	3597	0.384414
26	CHANGED_TOGETHER_WITH	3034	0.324246
27	ANNOTATED_BY	2924	0.312490
28	REQUIRES	2233	0.238643
29	HAS_FIRST_CHILD	2162	0.231055

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

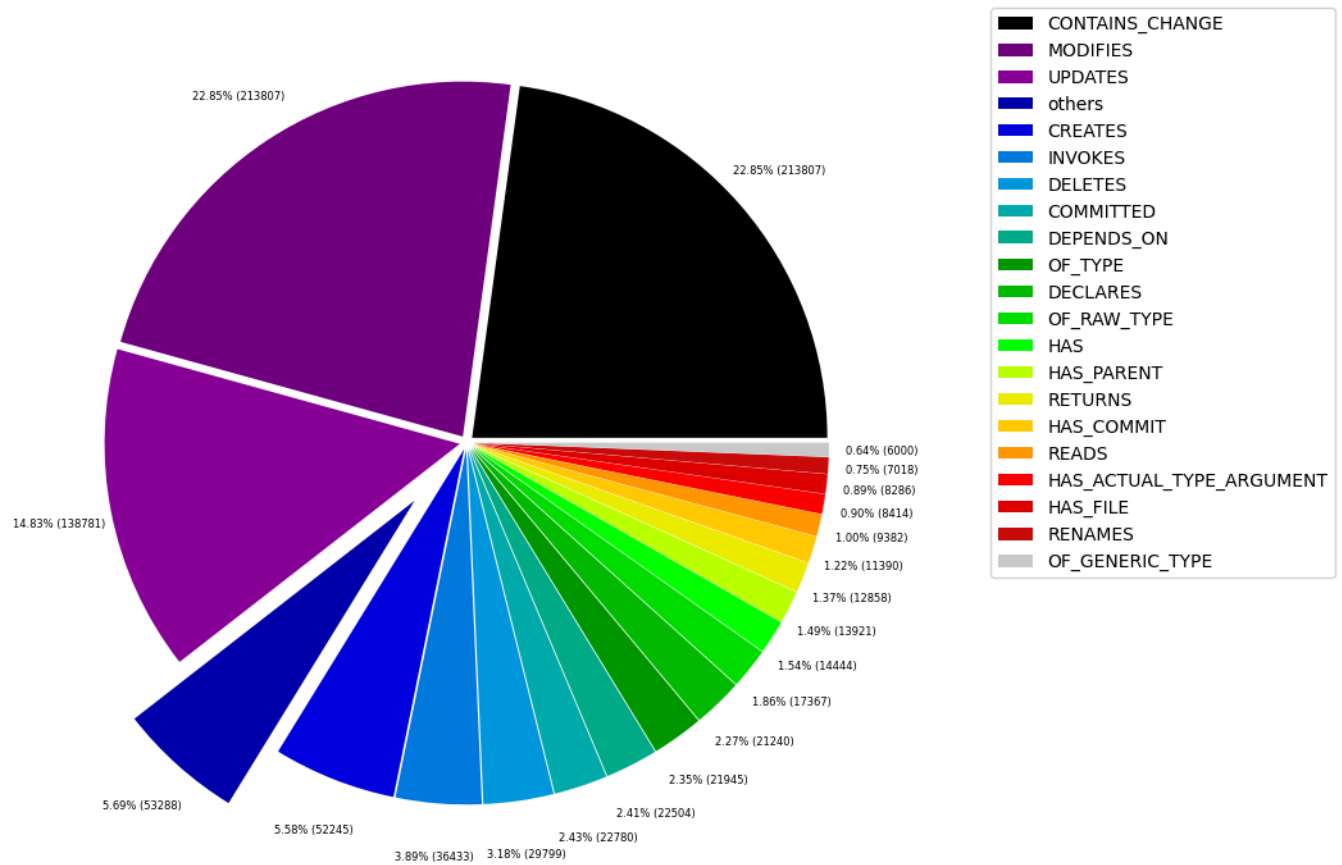


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

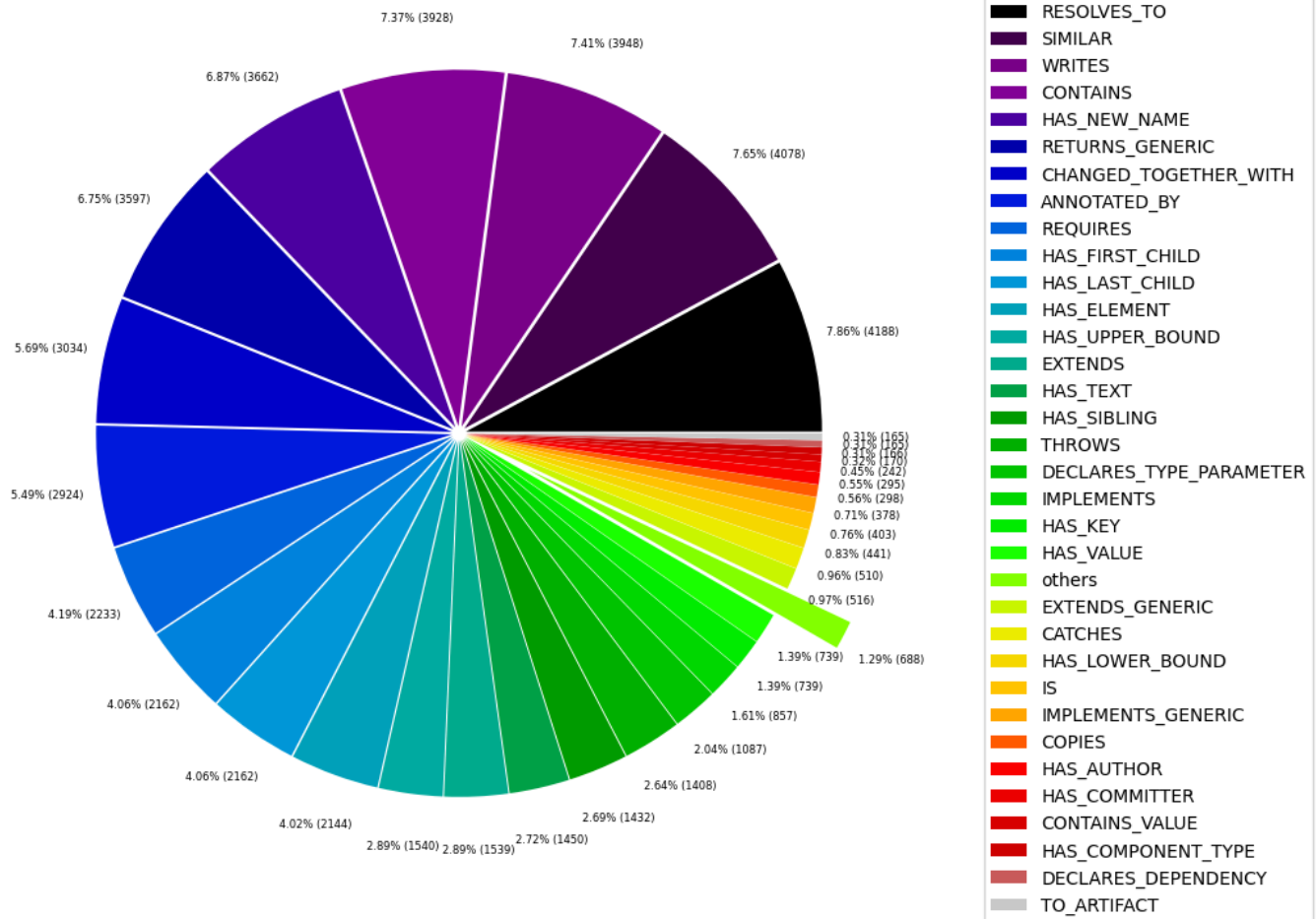
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000534
4	EXCLUDES	5	0.000534
5	DESCRIBES	9	0.000962
6	HAS_ROOT_ELEMENT	11	0.001176
7	HAS_GOAL	16	0.001710
8	HAS_EXECUTION	16	0.001710
9	OF_NAMESPACE	18	0.001924
10	HAS_ATTRIBUTE	18	0.001924
11	INCLUDES_CONCEPT	19	0.002031
12	USES_PLUGIN	21	0.002244
13	IS_ARTIFACT	21	0.002244
14	HAS_CONFIGURATION	21	0.002244
15	REQUIRES_TYPE_PARAMETER	24	0.002565
16	REQUIRES_CONCEPT	28	0.002992
17	DECLARES_NAMESPACE	36	0.003847
18	HAS_DEFAULT	39	0.004168
19	COPY_OF	119	0.012718
20	HAS_TAG	129	0.013786
21	ON_COMMIT	129	0.013786
22	DECLARES_DEPENDENCY	165	0.017634
23	TO_ARTIFACT	165	0.017634
24	HAS_COMPONENT_TYPE	166	0.017741
25	CONTAINS_VALUE	170	0.018168
26	HAS_COMMITTER	242	0.025863
27	HAS_AUTHOR	295	0.031527
28	COPIES	298	0.031848
29	IMPLEMENTS_GENERIC	378	0.040397

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	UPDATES	[File, Git]	138781	138781	
1	[Git, Update, Change]	MODIFIES	[File, Git]	138781	138781	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	138781	11390	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44929	11390	
4	[Git, Change, Create]	MODIFIES	[File, Git]	44929	44929	
5	[Git, Change, Create]	CREATES	[File, Git]	44929	44929	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22781	11390	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	22781	22781	
8	[Git, Change, Delete]	DELETES	[File, Git]	22781	22781	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	21983	13413	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13912	11390	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11390	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	11390	295	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11390	242	
14	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8482	13413	
15	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8457	13413	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8286	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11390	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6160	13306	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8286	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3466	7245	
25	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3114	7286	
26	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7286	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2698	13306	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2520	2109	
29	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2475	7286	

Graph Density

```
total_number_of_nodes (vertices): 301206
total_number_of_relationships (edges): 935709
-> total directed graph density: 1.0313712263596311e-05
-> total directed graph density in percent: 0.0010313712263596312
```